

Reliable Leak Detection in Water Distribution Networks using a Lightweight Physics-Aware Artificial Neural Network

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Abstract

Water Distribution Networks (WDNs) face water loss from undetected leaks, resulting in major economic losses. While deep learning architectures like transformers achieve high benchmarks, they often operate as **black boxes** that lack physical consistency and require resources to capture network dynamics. Our study proposes a lightweight model, the **Physics-Aware Artificial Neural Network** (PA-ANN), that embeds hydraulic conservation laws directly into the loss function and training objective. By enforcing these equations, the model ensures leak detection remains consistent with mass and energy conservation, a critical requirement for engineering trust. Validated on the LeakDB Hanoi dataset, the proposed model scores **96.59%** precision and an F1 score of **85.98%**. Although Transformer models achieve higher metrics, our model uses **770k parameters** compared to millions used by transformers, allowing deployment on edge and IoT devices. The model also achieves a **62.98%** reduction in false positive alarms over the base ANN model, resulting in a reduction in operational costs and human effort. In parallel, a rule-based localization system achieves **79.17%** accuracy in finding the leak node, while reducing average fault isolation time by **63.6%** over the base model. These results demonstrate that our proposed approach provides a robust, reliable, and efficient solution for real-time urban monitoring.

Keywords: Environmental Sustainability, Hydraulic Constraint Modeling, Leak Detection, Physics-aware Neural Networks, Time-series Classification, Urban Water infrastructure

Highlights

- Proposes a lightweight physics-aware ANN (PA-ANN) for leak detection in water distribution networks.
- Targets false alarm reduction as a key bottleneck in practical leak monitoring.
- Embeds hydraulic conservation laws to improve reliability and physical consistency.
- Achieves 96.59% precision and 62.98% reduction in false alarms vs standard ANN.
- Supports scalable monitoring across network topologies.

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1. Introduction

Water Distribution Networks (WDNs) are among the most vital infrastructures, but with increasing population density and rapid urbanization, the rise in WDN complexity has also led to an increase in leakages [18] [5], with some places suffering as much as 70-80% water loss due to leakages [19], amounting to an estimated \$39 billion in non-revenue water losses globally [10]. Traditionally, leak detection has relied on manual inspection or acoustic monitoring, but with the availability of SCADA data and advancements in machine learning, a shift toward data-driven methods has been observed. Advanced models such as CNNs, LSTMs, GNNs, and Transformers have achieved impressive accuracies and theoretical performance, but their adoption in real-world urban monitoring remains limited. The primary reason is the **trust gap** due to the **black-box** nature of these models. In critical infrastructure, it is important that predictions are not only correct but also physically consistent with the underlying hydraulic principles. Moreover, the computational cost and time required by these models are also barriers to their deployment on edge devices and IoT sensors. Another major problem is that current solutions prioritize finding all leaks (recall) over lowering false positive alarms (precision). In a real system, a leak is a rarity compared to the normal state; hence, each false alarm incurs significant human and monetary costs in inspections.

To address these challenges, this study proposes a lightweight **Physics-Aware Artificial Neural Network (PA-ANN)**. Our model incorporates hydraulic conservation laws, specifically the Equation of Continuity and the Hazen-Williams equation, into the loss function, providing a soft constraint on model predictions. Using only $\sim 770,000$ parameters, we achieve a **62.98%** reduction in false alarms compared to the standard ANN. A rule-based localization system also achieves **79.17%** accuracy in leak node localization, with PA-ANN reducing leak detection time by **63.6%** compared to ANN, providing a reliable and efficient system for practical use in urban monitoring.

2. Related work

Leak detection and localization in WDNs have been investigated through various methods, ranging from traditional data-driven machine learning to physics-informed approaches and hybrid/model-based methodologies. This section reviews recent advances, highlighting the research gap that motivates the present study.

2.1. Traditional Data-Driven Approaches

Traditional machine and deep learning show considerable promise by learning patterns from historical data. For instance, [14] used statistical features from sensor measurements with ANNs to achieve high detection reliability through fusion. Similarly, [20] demonstrated clustering strategies like K-means and Learning Vector Quantization (LVQ), which can be highly robust to common sensor noise. Deep learning architectures, such as Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks, have also been effective at capturing temporal dependencies [16, 1]. However, despite their high accuracy, these purely empirical traditional models face a critical limitation: they require extensively labeled data, struggle with generalization under expected distribution shifts, and inherently lack physical interpretability.

2.2. Hybrid and Model-Based Methodologies

Recognizing the limitations of purely data-driven methods, recent research has integrated hydraulic modeling with machine learning [6]. Reference [15] proposed a hybrid framework combining EPANET hydraulic simulations with bias correction and logistic regression, localizing leaks while handling model-data discrepancies. Other advancements focus on architectural improvements; [13] utilized a novel transformer model that concatenates position embeddings to preserve subtle anomalous features in historical pressure data. Additionally, [2] introduced a multilayer network representation integrating topological and sensor graphs with statistical anomaly detection. This regionalized approach reduced the need for exhaustive hydraulic simulations but struggled with incipient leaks exhibiting gradual flow increases, emphasizing the need to continuously enforce physical constraints.

2.3. Physics-Informed Neural Networks *in* WDNs

Physics-Informed Neural Networks (PINNs) represent an emerging paradigm that embeds governing partial differential equations directly into the neural network’s loss function, ensuring adherence to physical laws [17]. Reference [21] pioneered this in WDNs, using a PINN-based deep autoencoder incorporating continuity and momentum equations to accurately localize leaks without requiring a calibrated hydraulic model. Expanding on this foundation, [4] trained surrogate PINN models on Navier-Stokes equations to successfully detect sensor-based cyberattacks, but in small-scale simulated environments. More recently, [11] developed a PINN-driven digital twin framework using statistical control charts on reconstruction errors for rapid real-world leak detection. While successful, this framework’s dependency on accurate boundary conditions and limited handling of transient flow regimes highlight ongoing challenges. Ultimately, while PINNs offer a robust bridge between physical laws and deep learning, addressing incipient leaks and transient dynamics remains an open area of investigation.

2.4. Physics-Informed Neural Networks (PINNs)

Physics-Informed Neural Networks (PINNs) [17] are designed to solve **continuous partial differential equations (PDEs)** that govern physical systems. Unlike standard black-box models, PINNs embed the underlying physical laws directly into the loss function. This is achieved using **automatic differentiation** to compute exact derivatives of the network’s output with respect to its input coordinates. The total loss incorporates a PDE residual term evaluated at specific collocation points, effectively constraining the model to obey the laws of physics across the entire continuous domain.

2.5. Physics-Aware Neural Networks (PA-NNs)

In contrast, our approach uses **Physics-Aware Neural Networks**, which are better suited for discrete time-series data. Rather than solving a PDE, the model predicts physical quantities alongside the primary objective. These predictions are then used to compute the outputs of specific equations, which are added to the primary model loss. By treating physics as a soft constraint on discrete measurements, PA-ANNs achieve superior performance over baseline models while using significantly fewer parameters and resources compared to PINNs.

3. Dataset

3.1. Dataset Description

This study uses the **LeakDB Hanoi** dataset introduced by [8], which is made using the *Water Network Resilience Tool (WNTR)*. The dataset consists of a total of 999 scenarios, where each scenario is a different simulation of the **Hanoi** WDN for a year. The data points in each scenario are thirty minutes apart, which means there are 17,520 data points in one scenario, and a total of 17,502,480 data points in the entire dataset.

The Hanoi WDN, as shown in Figure 1, consists of 32 nodes and 34 pipes connecting them, and the LeakDB dataset assumes pressure, demand, and flow-rate sensors at these 32 nodes. Thus, each data point has 32 demand readings, 32 pipe readings, and 34 flow-rate readings, making a total of 98 values per time step.

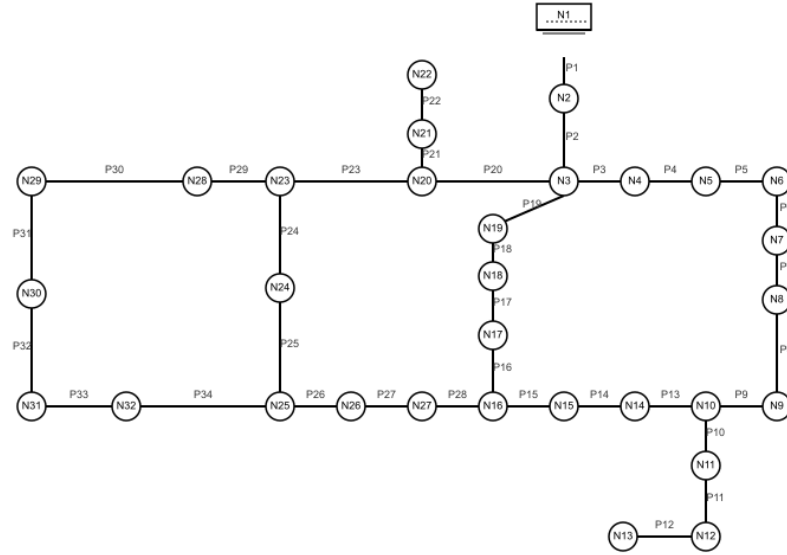


Figure 1: Figure of Hanoi WDN

The data can be downloaded using the `water_benchmark_hub` Python module. There are two types of readings available, namely `raw_data` and `scada_data`. In this work, `raw_data` is used as it also includes demand readings.

Initially, each scenario is stored as a directory containing subfolders for *demand*, *pressure*, *flow*, and *leaks*, where leak labels are multi-class values ranging from 0 to 32. In addition, each scenario includes an *INP* file, which specifies the simulated length, diameter, and roughness coefficient of each pipe. For the learning task, the dataset is reformatted into a single `.csv` file per scenario, where leak labels are converted to a binary representation (leak / non-leak).

Leak localization is not learned directly by the model and is instead performed as a post-processing step using a rule-based system.

3.2. Exploratory Analysis of Data

As shown in Figure 2, the heatmap below summarizes all 999 scenarios with 5% of rows from each scenario selected randomly. The demand and pressure readings show a constant negative correlation, reflecting the drop in pressure when demand increases, and vice versa. Flow readings show mixed correlations with demand and pressure, indicating the directional and topological effects of the network. These correlations support the use of both dense and physics-aware models.

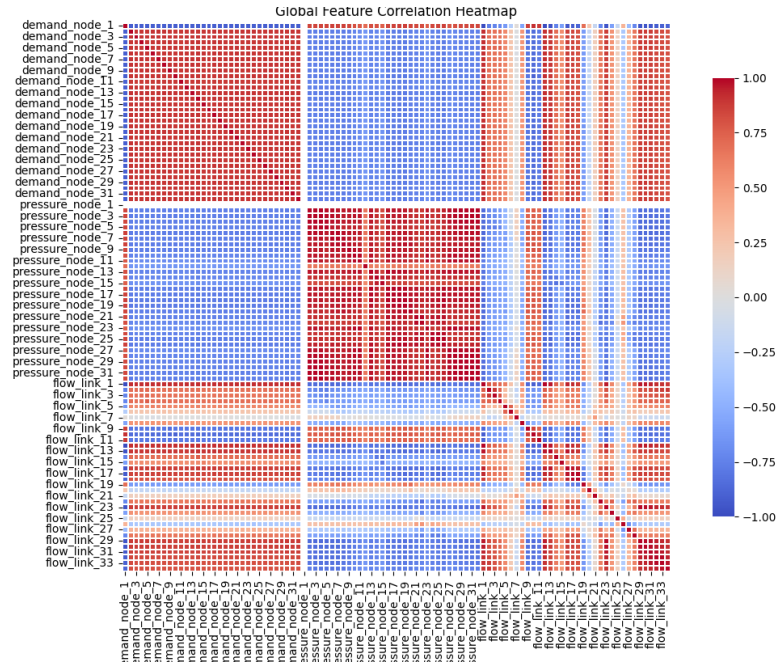


Figure 2: Correlation heatmap of Hanoi WDN

In the dataset, as shown in Figure 3, 4,311,716 rows have a leak state, while 13,190,764 rows have a non-leak state. This means that the ratio of non-leak data points to leak points is **3.059**, making the dataset highly imbalanced and challenging for any neural network.

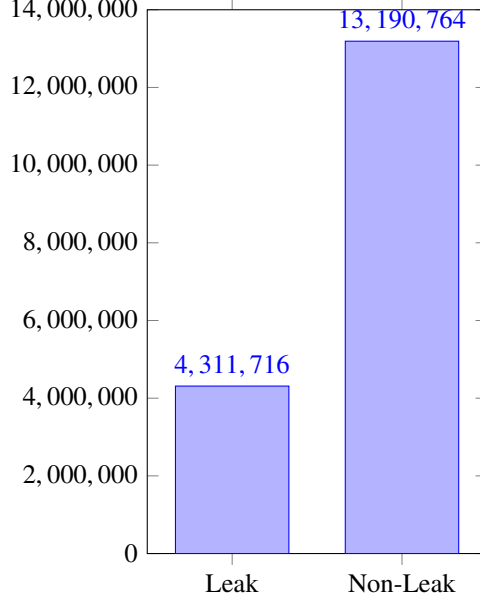


Figure 3: Distribution of leak and non-leak rows in the dataset.

A scenario that has at least one leak state can be called a *leak scenario*, while a scenario that has no leak state at any timestamp can be called a *non_leak scenario*. Out of the 999 scenarios, 758 are *leak scenarios*, while 241 are *non_leak scenarios*. This ratio is also maintained while selecting scenarios for training, validation, and testing.

3.3. Feature Engineering and Pre-Processing

In terms of feature engineering, two features are added to each scenario to increase model performance. As the data does not have any missing or *NaN* values, filling in missing data is not needed, and only data normalization is performed during pre-processing.

3.3.1. Adding columns

Pressure, demand, and flow in WDNs exhibit strong diurnal behavior: daytime operation typically shows higher demand with corresponding hydraulic shifts, while nighttime behavior follows the opposite pattern. Without explicit time-of-day information, the model can confuse regular daily fluctuations with leak signatures. To capture this periodic context, two cyclical temporal features, *sin_hour* and *cos_hour*, are added to every scenario.

Let i denote the row index within a scenario. The corresponding *step of day* s is defined as:

$$s_i = i \bmod 48$$

Using this discrete step index, the temporal features are encoded as:

$$\sin_hour_i = \sin\left(\frac{2\pi s_i}{48}\right)$$

$$\cos_hour_i = \cos\left(\frac{2\pi s_i}{48}\right)$$

These paired sinusoidal features preserve the circular nature of time and help the classifier distinguish normal diurnal variation from anomalous behavior. For example, 12 A.M. ($\sin_hour = 0$ and $\cos_hour = 1$) and 9 A.M. ($\sin_hour = 0.707$ and $\cos_hour = -0.707$) receive different temporal encodings even if pressure, demand, and flow values are otherwise similar.

3.3.2. Normalization

All sets (train, validation, and test) are normalized using the mean and standard deviation of the training set, such that the mean of each set is 0 and the standard deviation is 1, after normalization.

Let $\mathcal{D}_{\text{train}} = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N\}$ denote the training dataset, where each feature vector $\mathbf{x}_i \in \mathbb{R}^F$.

The global mean and standard deviation are computed feature-wise over the training set as:

$$\boldsymbol{\mu} = \frac{1}{N} \sum_{i=1}^N \mathbf{x}_i, \quad (1)$$

$$\boldsymbol{\sigma} = \sqrt{\frac{1}{N} \sum_{i=1}^N \mathbf{x}_i^{\circ 2} - \boldsymbol{\mu}^{\circ 2}}, \quad (2)$$

where $\circ$ denotes element-wise operations.

To ensure numerical stability, the variance is lower bounded by a small constant ϵ :

$$\boldsymbol{\sigma} = \sqrt{\max\left(\frac{1}{N} \sum_{i=1}^N \mathbf{x}_i^{\circ 2} - \boldsymbol{\mu}^{\circ 2}, \epsilon\right)}. \quad (3)$$

Using the statistics computed from the training set, each feature is normalized as:

$$\tilde{\mathbf{x}} = \frac{\mathbf{x} - \boldsymbol{\mu}}{\boldsymbol{\sigma}}, \quad (4)$$

where $\tilde{\mathbf{x}}$ denotes the normalized feature vector.

The same training-set mean $\boldsymbol{\mu}$ and standard deviation $\boldsymbol{\sigma}$ are applied consistently to the training, validation, and test sets to avoid data leakage.

3.4. Windowing function

To give the model temporal context and features without using recurrent architectures such as LSTMs or GRUs, a sliding-window-based dataset construction is used. Temporal windows are extracted from each scenario using a fixed window size of 12, or 6 hours, as each time step is 30 minutes. For each window, the target is the leak state (0/1) at the final timestamp of the window, allowing the model to learn temporal patterns preceding a leak event. As the target label is the leak state at the final timestamp, which is also the current timestamp, this means that the model **nowcasts** leak predictions.

For the ANN models, each window is flattened into a one-dimensional vector as input to the network. A window of 12 timestamps with 100 features per timestamp is reshaped into a one-dimensional vector of length 12×100 . The mathematical representation is:

Let a scenario be represented as a multivariate time series

$$\mathbf{X} = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_T\}, \quad \mathbf{x}_t \in \mathbb{R}^F, \quad (5)$$

where F denotes the number of features at each time step.

Using a sliding window of length W , the k -th temporal window is defined as

$$\mathbf{X}^{(k)} = [\mathbf{x}_k, \mathbf{x}_{k+1}, \dots, \mathbf{x}_{k+W-1}] \in \mathbb{R}^{W \times F}. \quad (6)$$

For the ANN model, each window is reshaped into a one-dimensional feature vector via vectorization:

$$\tilde{\mathbf{x}}^{(k)} = \text{vec}(\mathbf{X}^{(k)}) \in \mathbb{R}^{W \cdot F}. \quad (7)$$

This flattened representation serves as the input to the fully connected network.

4. Physics Loss

All physical systems are governed by the conservation of mass and conservation of energy, and the Water Distribution Network is also assumed to be governed by these laws. The network is modeled as a directed graph, where vertices represent nodes, and edges represent pipes connecting nodes. *Mass conservation* is enforced at each node, and *Energy conservation* is enforced along each pipe. The equations used for these are discussed below.

4.1. Equations

For *Mass conservation*, the **Continuity Equation** is used. This means that the inflow at each node is equal to the sum of the demand at that node and the outflow at each node.

$$r_{\text{mass}}^{(i)} = \sum_{j \in \mathcal{E}_{\text{in}}(i)} Q_j - \sum_{j \in \mathcal{E}_{\text{out}}(i)} Q_j - D_i,$$

where Q_j denotes the flow through pipe j , D_i is the demand, and $\mathcal{E}_{\text{in}}(i)$ and $\mathcal{E}_{\text{out}}(i)$ represent incoming and outgoing pipes, respectively.

For *Energy conservation*, the *Hazen-Williams Equation* is used. The **Hazen-Williams Equation** defines the loss due to friction in a pipe; this frictional loss, along with the elevation head differences between the two ends of a pipe, gives the pressure drop. Since in the LeakDB dataset, in all scenarios, all nodes are kept at the same elevation, the pressure drop due to elevation differences is assumed to be negligible. The Hazen-Williams head loss for pipe k is given by

$$h_k = 10.67 \frac{L_k |Q_k|^{1.852}}{C_k^{1.852} D_k^{4.87}},$$

where L_k is the pipe length, D_k is the diameter, C_k is the roughness coefficient, and Q_k is the predicted flow.

4.2. Physics Loss Algorithm

The physics loss due to mass conservation is defined as:

$$\mathcal{L}_{\text{mass}} = \frac{1}{N} \sum_{i=1}^N \left(\frac{r_{\text{mass}}^{(i)}}{\bar{D}} \right)^2,$$

where $\bar{D}$ denotes a scale factor (e.g., mean absolute demand) for numerical stability.

The predicted pressure drop across a pipe is given as:

$$\Delta p_k = \text{sign}(Q_k) (p_{\text{src}(k)} - p_{\text{dst}(k)}),$$

where $p_{\text{src}(k)}$ and $p_{\text{dst}(k)}$ are the predicted pressures at the source and destination nodes.

Thus, the loss due to energy conservation is defined as:

$$r_{\text{energy}}^{(k)} = \Delta p_k - h_k,$$

$$\mathcal{L}_{\text{energy}} = \frac{1}{M} \sum_{k=1}^M \left(\frac{r_{\text{energy}}^{(k)}}{\bar{p}} \right)^2,$$

where $\bar{p}$ denotes a scale factor (e.g., mean absolute predicted pressure) for numerical stability.

The total physics loss is a weighted average of the loss due to mass conservation and energy conservation, given as:

$$\mathcal{L}_{\text{phys}} = \lambda_{\text{mass}} \mathcal{L}_{\text{mass}} + \lambda_{\text{energy}} \mathcal{L}_{\text{energy}},$$

where λ_{mass} and λ_{energy} are hyperparameters; in our implementation we use $\lambda_{\text{mass}} = 1.0$ and $\lambda_{\text{energy}} = 0.1$ to emphasize mass conservation consistency.

4.3. Pseudocode and Training Integration

Algorithm 1 Physics Loss Computation

Require: $\hat{p}$ (pred. pressure), $\hat{Q}$ (pred. flow), D (demand), L, D, C (pipe params), edge_index

Ensure: $\mathcal{L}_{\text{phys}}, \mathcal{L}_{\text{total}}$

- 1: **Aggregate Flow at Nodes**
 - 2: $\text{flow_in}[i] \leftarrow \sum_{j:\text{dst}(j)=i} \hat{Q}_j$
 - 3: $\text{flow_out}[i] \leftarrow \sum_{j:\text{src}(j)=i} \hat{Q}_j$
 - 4: **Mass Conservation Residual**
 - 5: $r_{\text{mass}}^{(i)} \leftarrow \text{flow_in}[i] - \text{flow_out}[i] - D_i$
 - 6: $\mathcal{L}_{\text{mass}} \leftarrow \text{mean}\left(\left(\frac{r_{\text{mass}}}{D}\right)^2\right)$
 - 7: **Hazen-Williams Head Loss**
 - 8: $h_k \leftarrow 10.67 \cdot L_k \frac{|\hat{Q}_k|^{1.852}}{C_k^{1.852} D_k^{4.87}}$
 - 9: **Predicted Pressure Drop**
 - 10: $\Delta p_k \leftarrow \text{sign}(\hat{Q}_k)(\hat{p}_{\text{src}(k)} - \hat{p}_{\text{dst}(k)})$
 - 11: **Energy Residual**
 - 12: $r_{\text{energy}}^{(k)} \leftarrow \Delta p_k - h_k$
 - 13: $\mathcal{L}_{\text{energy}} \leftarrow \text{mean}\left(\left(\frac{r_{\text{energy}}}{\bar{p}}\right)^2\right)$
 - 14: **Total Physics Loss**
 - 15: $\mathcal{L}_{\text{phys}} \leftarrow \lambda_{\text{mass}} \mathcal{L}_{\text{mass}} + \lambda_{\text{energy}} \mathcal{L}_{\text{energy}}$
 - 16: **Combined Training Loss**
 - 17: $\mathcal{L}_{\text{total}} \leftarrow \mathcal{L}_{\text{WBCE}} + \lambda_{\text{phys}} \mathcal{L}_{\text{phys}}$
-

Algorithm 2 Training Loop with Physics Loss

Require: Training data $(x_b, y_b, \text{scenario_ids}, D_b)$

Ensure: Updated model parameters

Set Physics Weight by Epoch

```
2: if  $e < 10$  then
     $\lambda_{\text{phys}} \leftarrow 0.01$ 
4: else if  $10 \leq e < 20$  then
     $\lambda_{\text{phys}} \leftarrow 0.03$ 
6: else
     $\lambda_{\text{phys}} \leftarrow 0.05$ 
8: end if
for each batch  $(x_b, y_b, \text{scenario\_ids}, D_b)$  do
10:   Load pipe parameters  $(L, D, C)$  for batch
    Denormalize  $D_b$  to physical units
12:    $(\hat{y}, \hat{p}, \hat{Q}) \leftarrow \text{model}(x_b)$ 
    Denormalize  $\hat{p}, \hat{Q}$ 
14:    $\mathcal{L}_{\text{WBCE}} \leftarrow \text{weighted\_BCE}(\hat{y}, y_b)$ 
    Compute  $\mathcal{L}_{\text{total}}$  using Algorithm 1
16:   Backpropagate  $\mathcal{L}_{\text{total}}$ 
    Update model parameters
18: end for
```

4.4. Integration into Model

To use this loss in the ANN model, the *.inp* files of each scenario are formatted into a *.csv* file containing the pipe length, diameter, and roughness coefficient of each pipe in the scenario, and this information is integrated into the dataset class. As the node-pipe connection is constant for all scenarios, a fixed *edge_list* is used. During training, the physics loss is added to the classification loss, and the result is used for backpropagation.

5. Model Architecture and Training Details

5.1. Architecture

5.1.1. Base ANN Model

The model here is a fully connected feed-forward network designed for classification of the presence of a leak at a single timestamp. The input layer consists of 1200 features due to the windowing function. It is passed through three hidden layers of 512, 256, and 64 neurons. The first two hidden layers employ batch normalization and ReLU activation, and a dropout layer (rate = 0.5) to prevent overfitting. The third hidden layer uses only ReLU activation. The output layer consists of a single neuron with a sigmoid activation function for binary leak classification.

ANN Loss:

As discussed above, because the model is a binary classification model, and the class imbalance is severe, **Class-weighted binary cross-entropy** loss is used, where the loss is reweighted inversely proportional to class frequency.

Let N_0 denote the number of non-leak samples, and let N_1 denote the number of leak samples.

$$w_0 = \frac{N_0 + N_1}{2N_0}, \quad w_1 = \frac{N_0 + N_1}{2N_1}$$

Let $y_i \in \{0, 1\}$ denote the ground-truth leak label for the i -th sample, and let $\hat{y}_i \in (0, 1)$ denote the corresponding predicted leak probability.

$$\mathcal{L}_{\text{WBCE}} = -\frac{1}{N} \sum_{i=1}^N [w_1 y_i \log(\hat{y}_i) + w_0 (1 - y_i) \log(1 - \hat{y}_i)]$$

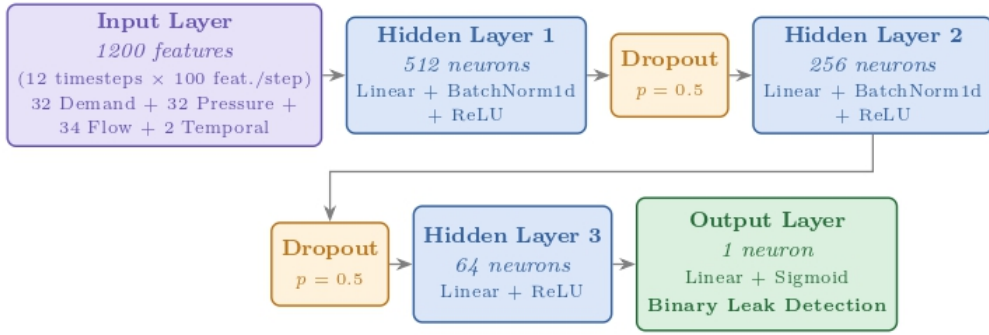


Figure 4: Architecture Diagram of ANN

5.1.2. Physics-Aware ANN Model

The model here is a fully connected feed-forward network designed for pressure and flow prediction, and classification of the presence of a leak at a single timestamp, using those predictions. The input layer is 1200 features here as well. It is passed through three hidden layers of 512, 256, and 64 neurons. All three hidden layers employ batch normalization, ReLU activation, and a dropout layer (rate = 0.5). The third hidden layer is connected to two layers, one with 32 neurons and one with 34 neurons, to predict pressure and flow, respectively. The leak head consists of two layers, with the input layer being the 64 neurons of the third hidden layer and five features derived from the predicted pressure and flow (mean of pressure and flow, standard deviation of pressure and flow, and the range of pressure), thus 69 features in total. These are then passed through a hidden layer of 64 neurons, where ReLU activation and a dropout layer are employed, and the output layer is a single neuron with a sigmoid activation function for binary leak classification.

Physics-Aware ANN Loss:

The **Class-weighted binary cross-entropy**, also the **Classification loss** of the model is given by:

Let N_0 denote the number of non-leak samples, and let N_1 denote the number of leak samples.

$$w_0 = \frac{N_0 + N_1}{2N_0}, \quad w_1 = \frac{N_0 + N_1}{2N_1}$$

Let $y_i \in \{0, 1\}$ denote the ground-truth leak label for the i -th sample, and let $\hat{y}_i \in (0, 1)$ denote the corresponding predicted leak probability.

$$\mathcal{L}_{\text{WBCE}} = -\frac{1}{N} \sum_{i=1}^N [w_1 y_i \log(\hat{y}_i) + w_0 (1 - y_i) \log(1 - \hat{y}_i)]$$

Let the physics loss, as discussed above in detail above, be represented as: $\mathcal{L}_{\text{phys}}$
Thus, the total loss of the model is given by:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{WBCE}} + \lambda_{\text{phys}} \mathcal{L}_{\text{phys}}.$$

The λ_{phys} follows this schedule:

$$\lambda_{\text{phys}}(e) = \begin{cases} 0.01, & e < 10, \\ 0.03, & 10 \leq e < 20, \\ 0.05, & 20 \leq e < 30, \end{cases}$$

This is done to make sure that the physics loss does not dominate the classification loss at the start.

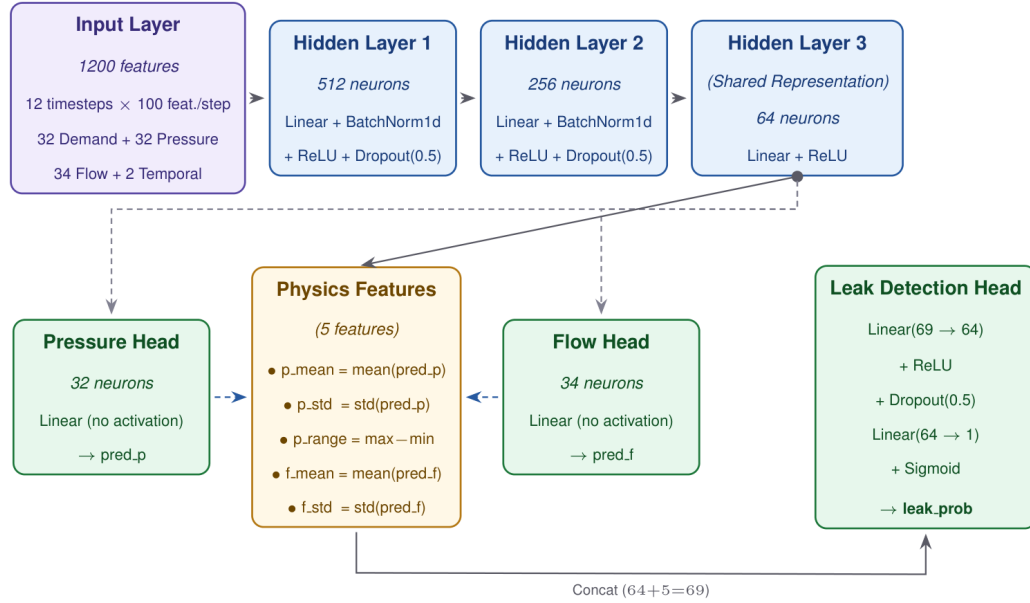


Figure 5: Architecture Diagram of Physics-Aware ANN

6. Localization

6.1. Introduction

While the deep learning (ANN) module handles the temporal detection of anomalies (identifying *when* a leak occurs), spatial localization (identifying *where* the leak is located) is managed

by a topology-informed statistical engine. Designed specifically for real-time operation, this algorithm processes sequential data streams. This hybrid architecture leverages the strengths of both domains: the ANN captures complex, non-linear patterns to detect incipient events, while the localization module relies on hydraulic first principles and network graph theory.

A major challenge in hydraulic localization is signal drift. When a leak occurs, the initial pressure drop is most severe at the source node. However, as the transient wave propagates, pressure drops accumulate at physical boundaries, such as dead-ends or downstream junctions, often causing purely statistical algorithms to drift away from the true source over time. To combat this, the proposed real-time module introduces an "Anti-Drift" mechanism combining time-decayed voting, topology-based neighbor validation, and a stability lock, triggered only when the ANN confidence exceeds a threshold ($\tau_{conf} > 0.75$).

6.2. Algorithm

The real-time localization algorithm operates in three distinct phases: Lagged Baseline Snapshot, Topological Scoring, and Time-Decayed Voting.

6.2.1. Lagged Baseline Snapshot

During normal operation (when the ANN probability $P(leak) < 0.60$), the system maintains a lightweight rolling buffer of pressure readings for all N sensor nodes. Upon ANN triggering a leak state at time $t_{trigger}$, the system immediately captures a "snapshot" of the network's healthy state.

To prevent the baseline from being contaminated by the incipient, pre-trigger transient wave of the leak, a lag period ($W_{lag} = 50$ steps) is excluded. The reference mean (μ_i) and standard deviation (σ_i) for each sensor i are calculated from a stable historical window ($W_{base} = 200$ steps) preceding the lag:

$$\mathcal{B}_{clean} = P_{t_{trigger}-W_{lag}-W_{base} : t_{trigger}-W_{lag}} \quad (8)$$

This frozen baseline acts as the static reference frame for the duration of the event.

6.2.2. Topological and Physics-Aware Scoring

At each active leak step t , the algorithm calculates the positive pressure residual and standardizes it into a Z-score, applying a noise floor ($\epsilon = 0.05$):

$$Z_i(t) = \frac{\max(0, \mu_i - p_i(t))}{\max(\sigma_i, \epsilon)} \quad (9)$$

Rather than evaluating nodes in isolation, the system utilizes the network's adjacency graph. A node i is evaluated based on its own drop, the supporting drops of its physical neighbors $\mathcal{N}_i$, and its structural degree D_i .

If $Z_i(t) > 1.0$, a Topological Score $S_i(t)$ is computed. It incorporates a Neighbor Support Ratio (R_i), rewarding nodes whose adjacent neighbors also exhibit drops, and a Degree Multiplier ($M_i = 1 + 0.05 \times D_i$), which lightly penalizes dead-ends ($D = 1$) in favor of central junctions:

$$R_i = \min \left(1.0, \frac{\frac{1}{|\mathcal{N}_i|} \sum_{j \in \mathcal{N}_i} Z_j(t)}{Z_i(t)} \right) \quad (10)$$

$$S_i(t) = Z_i(t) \times (1 + R_i) \times M_i \quad (11)$$

6.2.3. Time-Decayed Voting and Anti-Drift Lock

To prioritize the initial hydraulic wave closest to the true leak source, the system applies a rapid time-decaying weight $W(t)$ based on the duration $\Delta t = t - t_{trigger}$:

$$W(t) = \frac{1}{1 + 0.1 \times \Delta t} \quad (12)$$

At each step, the top 3 candidate nodes are awarded points scaled by $W(t)$, and the cumulative vote V_i for each node is recursively updated.

To definitively prevent late-stage drift toward network boundaries, a **Stability Lock** is enforced. If the leading suspect $\hat{L}$ remains unchanged for a consecutive threshold of $k_{lock} = 10$ time steps, the algorithm considers the localization converged and halts suspect updating.

Handling Simultaneous Leaks: In scenarios involving multiple simultaneous bursts, the combination of time-decayed voting and the Stability Lock acts as a dominant-signal isolator. The node experiencing the most severe and immediate pressure gradient (the primary leak) accumulates early votes rapidly. The algorithm intentionally locks onto this primary source, preventing the system from oscillating confusingly between multiple coordinates. Secondary leaks can subsequently be localized once the primary anomaly is isolated by network operators and the baseline resets.

Algorithm 3 Real-Time Physics-Aware Localization Strategy

Require: Real-time pressure stream P_t , ANN model $\mathcal{M}$, adjacency graph $\mathcal{G}$

Ensure: Localized Node L_{est}

```
1:  $\mathcal{B} \leftarrow \emptyset$ ,  $V \leftarrow \vec{0}$ ,  $LeakActive \leftarrow \text{False}$ ,  $IsLocked \leftarrow \text{False}$ 
2:  $L_{curr} \leftarrow \text{null}$ ,  $Stability \leftarrow 0$ 
3: while stream is active at time  $t$  do
4:    $p_{leak} \leftarrow \mathcal{M}(P_{t-w:t})$ 
5:   if not  $LeakActive$  then
6:     if  $p_{leak} < 0.60$  then Append  $P_t$  to  $\mathcal{B}$ 
7:     end if
8:     if  $p_{leak} > 0.75$  then
9:        $LeakActive \leftarrow \text{True}$ ,  $t_{start} \leftarrow t$ 
10:       $\mu, \sigma \leftarrow \text{ExtractLaggedStats}(\mathcal{B}, lag = 50, win = 200)$ 
11:    end if
12:   else
13:     if  $p_{leak} < 0.40$  then Reset System State
14:     end if
15:     if not  $IsLocked$  then
16:        $W_t \leftarrow 1/(1 + 0.1 \times (t - t_{start}))$ 
17:        $C \leftarrow \emptyset$  ▷ Candidate List
18:       for each node  $i \in \{1, \dots, N\}$  do
19:          $Z_i \leftarrow \max(0, \mu_i - P_{i,t})/\sigma_i$ 
20:         if  $Z_i > 1.0$  then
21:           Add  $(i, \text{CalcTopoScore}(Z_i, \mathcal{G}))$  to  $C$ 
22:         end if
23:       end for
24:       Sort  $C$  descending by TopoScore
25:       for rank  $r \in \{0, 1, 2\}$  from top 3 of  $C$  do
26:          $V_{C[r].id} \leftarrow V_{C[r].id} + (3 - r) \times W_t$ 
27:       end for
28:        $L_{new} \leftarrow \text{argmax}(V)$ 
29:       if  $L_{new} == L_{curr}$  then
30:          $Stability \leftarrow Stability + 1$ 
31:         if  $Stability > 10$  then  $IsLocked \leftarrow \text{True}$ 
32:       end if
33:     else
34:        $L_{curr} \leftarrow L_{new}$ ,  $Stability \leftarrow 0$ 
35:     end if
36:   end if
37: end if
38: end while
```

7. Results and Discussion

Both models are trained on **490 scenarios**, with 105 scenarios used for validation and 105 scenarios used for testing. Both models are trained for 30 epochs, and for each model, the

checkpoint with the highest validation **F1** is used for testing. We also use an XGBoost model with 100 $n_estimators$. It is trained on the same windowed dataset as the neural networks.

Table 1: Comparison between ANN, PA-ANN and XGBoost on 105 scenarios

Model	Parameters	Acc.	Prec.	Rec.	F1
XGBoost	–	90.42%	95.67%	68.36%	79.74%
ANN	764,289	92.13%	91.43%	78.89%	84.70%
PA-ANN	773,187	93.03%	96.59%	77.47%	85.98%

Table 1 compares ANN and PA-ANN on 105 test scenarios from the Hanoi dataset. **PA-ANN** achieves higher accuracy, precision, and F1-score, while maintaining comparable recall. Notably, PA-ANN reduces false positives by approximately **63%** relative to ANN, making it significantly more reliable in practical deployment. This reduction directly translates to lower operational costs, as fewer false leak alarms reduce unnecessary inspections and interventions. The improvement is achieved without a substantial increase in model parameters, indicating that the gains stem from the incorporation of physics-based constraints rather than model complexity.

Table 2: Benchmark comparison

Model	Parameters	Accuracy	F1
Autoencoder [9]	~ 64,000	89%	78.84%
ANN	764,289	92.13%	84.70%
PA-ANN	773,187	93.03%	85.98%
CNN [16]	~ 1–1.2 M	88.89%	–
Transformer [12]	~ 1.5–2 M	95.19%	97.52%
LeakDB base [8]	–	26.8%	49.36%
GRU [3, 7]	–	90.68%	–

Table 2 benchmarks the proposed models against prior approaches. **PA-ANN** outperforms autoencoder and GRU-based methods while using fewer parameters than larger architectures. The CNN-based model reports lower accuracy despite significantly higher parameter counts, suggesting limited efficiency. Although the transformer model reports higher accuracy and F1-score, it is evaluated on only 9–10 scenarios and trained on a single scenario [12], making the comparison not directly equivalent to our much larger and more diverse **490/105 train-test setup**. This highlights that PA-ANN achieves competitive performance under significantly more realistic and challenging conditions. As the original thesis for the GRU-based method [3] was not directly accessible, its reported metrics are sourced from [7].

Table 3: Comparative Detection Performance on Net1 Dataset (100 Scenarios)

Model	Acc.	Prec.	Recall	F1
ANN	97.75%	93.98%	83.80%	88.60%
PA-ANN	98.10%	97.40%	84.12%	90.27%

Table 3 presents results on the Net1 dataset, consisting of **100 scenarios** (70 train, 15 validation, 15 test) sampled from a different water distribution network. **PA-ANN consistently outperforms**

ANN, achieving higher accuracy, precision, and F1-score. The improvement in precision again reflects a reduction in false alarms, reinforcing the conservative and reliable behavior of the physics-informed framework. These results demonstrate that the proposed approach generalizes effectively across different WDN configurations, rather than being tailored to a specific dataset.

7.1. Localization Results

Fault localization is performed using a deterministic rule-based decision system applied uniformly to outputs from both neural network models. The localization algorithm remains identical across experiments, ensuring that any observed performance differences arise solely from differences in the quality of model predictions.

Table 4: Fault localization performance comparison

Model	True Positives (TP)	False Negatives (FN)	Avg. Time to Localize
Physics-Aware ANN	95	25	11h 24m
Standard ANN	95	25	31h 19m
Autoencoder	-	-	13h 00m

Both models localize the same number of faults (TP = 95, FN = 25 out of 120 total cases), achieving identical localization accuracy of $\frac{95}{120} = 79.17\%$. This equality is expected because the rule-based spatial localization algorithm is robust to small variations in detection quality. However, the time required to localize differs dramatically:

$$\Delta T = 19h\ 55m \approx \mathbf{63.6\%} \text{ faster for Physics-Aware ANN}$$

This substantial improvement is driven by the 62.98% reduction in false positives at the detection level ($37,515 \rightarrow 13,887$ alarms), which results in fewer spurious signals entering the localization system. The physics-informed training produces outputs that align more efficiently with the rule-based decision logic, reducing ambiguity in leak signal patterns and enabling faster convergence to localization solutions. Thus, although both models achieve equal spatial accuracy, Physics-Aware ANN provides superior operational efficiency through more stable and reliable predictions.

8. Sensitivity Analysis

To assess model performance with changes in demand, pressure, and flow, we perform a **two-dimensional sensitivity analysis** over demand perturbation $\alpha \in [-0.15, 0.15]$ and pressure-demand coupling strength $\beta \in [0.2, 1.0]$.

The formulae are:

$$\text{Demand}_{\text{new}} = \text{Demand}(1 + \alpha)$$

$$\text{Flow}_{\text{new}} = \text{Flow}(1 + \alpha)$$

$$\text{Pressure}_{\text{new}} = \text{Pressure}(1 - \beta\alpha)$$

These formulae are used keeping in mind the correlations shown by the data heatmap.

Figure 6 shows the **worst-case F1** (minimum over all β), and we can see that for negative demand shifts, ANN degrades much more steeply compared to physics-aware ANN. At $\alpha = -0.15$, ANN drops to an F1 of 0.17, whereas physics-aware ANN holds at **0.31**. For positive demand shifts, both models show similar performance, and the metrics are almost constant for $\alpha \in [0.05, 0.15]$.

Figure 7 shows the performance of both models at an **extreme demand shift** ($\alpha = -0.15$), for varying β . As we can see, physics-aware ANN degrades gradually here, whereas the base ANN drops steeply from $\beta = 0.4$, where the F1 for both is approximately 0.7. At $\beta = 0.6$, physics-aware ANN shows an F1 of approximately **0.6**, while the base ANN drops to an F1 of approximately 0.4.

As the physics-aware ANN embeds **hydraulic constraints** during training, even under data shifts that may not strictly follow conservation of mass and energy, it maintains more stable predictions, whereas the base ANN degrades drastically due to being purely data-driven.

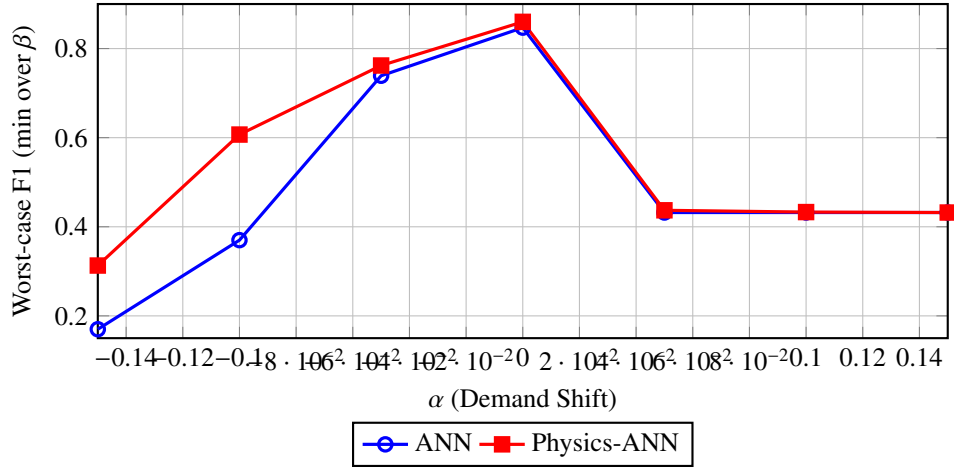


Figure 6: Worst case F1 (minimum over all β) across demand perturbations.

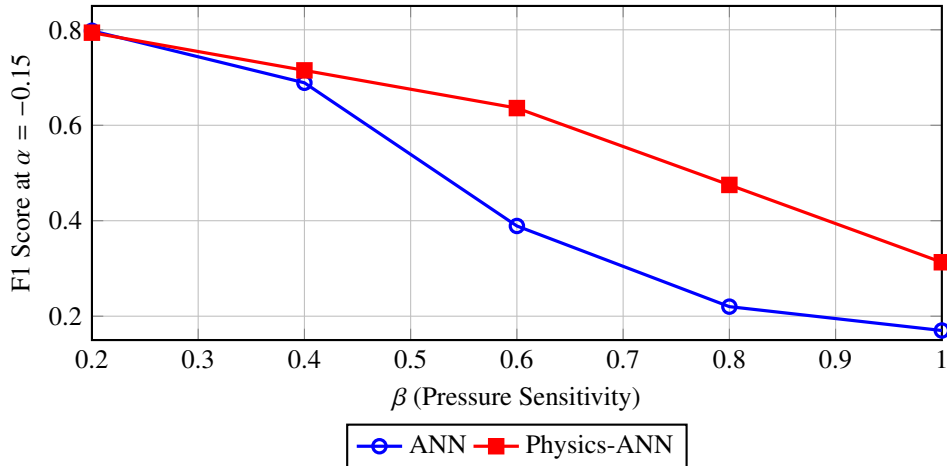


Figure 7: Performance of both models at an extreme demand shift ($\alpha = -0.15$) for varying β .

9. Limitations

The study has some limitations that must be acknowledged. First, cross-validation is not performed, and model evaluation is based on fixed train-validation-test splits. While the reported metrics are strong, additional validation strategies such as k-fold cross-validation can improve robustness assessment. Second, the proposed framework is evaluated on benchmark datasets (Hanoi and Net1). Although this demonstrates cross-network performance, results may not directly generalize to larger real-world distribution systems with different sensor quality, operating regimes, and demand dynamics. Future work can address these points through broader validation and deployment-oriented studies.

10. Conclusion

This work presents a **Physics-Aware ANN (PA-ANN)** for leak detection in water distribution networks, integrating hydraulic constraints into the learning process. The proposed model consistently outperforms a standard ANN by achieving higher precision and F1-score while reducing false positives by approximately **63%**, leading to more reliable and cost-effective leak detection. Across both the Hanoi and Net1 datasets, **PA-ANN demonstrates strong generalization**, maintaining performance improvements on a different network configuration. Unlike larger models such as transformers, which are evaluated under limited settings, PA-ANN achieves **competitive results** under a significantly broader and more realistic training and testing setup. Additionally, the physics-informed predictions improve downstream localization, reducing average fault isolation time by **over 60%** compared to ANN. This highlights that enforcing physical consistency not only enhances detection accuracy but also stabilizes predictions for faster decision-making. Overall, the results show that incorporating **domain knowledge** into neural networks provides a practical balance between performance, efficiency, and robustness, making PA-ANN suitable for **real-world deployment**.

11. Future Work

Natural extensions include applying the Physics-Aware ANN to additional LeakDB topologies beyond Hanoi and Net1 and to real-world SCADA datasets, incorporating EPANET-based perturbations in the sensitivity analysis, and exploring hybrid graph-physics architectures for larger networks. The rule-based localization component can be extended and benchmarked against alternative approaches. These directions build on the physics-aware framework presented here.

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Disclosure statement

The authors report no competing interests to declare.

Data availability statement

The data that support the findings of this study are publicly available. The LeakDB benchmark dataset and the Hanoi Water Distribution Network data are available from publicly accessible repositories such as <https://zenodo.org/records/13985057>. Additional processed data and implementation details are available from the authors upon reasonable request.

Author Contributions

Jash Shah, David Daniels, and Heet Shah contributed equally to this work and were involved in conceptualization, methodology, software development, data curation, formal analysis, and writing – original draft preparation.

Abhijeet Salunke contributed to supervision, validation, and critical review and editing of the manuscript.

All authors were involved in the interpretation of results, approved the final version of the manuscript, and agree to be accountable for all aspects of the work.

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Appendix

Ablation Studies

We present ablations on three decisions: **the window size**, **the temporal columns**, and **the physics-aware mechanism**. All ablations except the physics-aware mechanism are performed on the base ANN model trained on 70 scenarios and tested on 15 scenarios. The physics-aware mechanism is ablated on the Physics-Aware model trained and tested on the same scenarios.

Window-Size

The windowing function provides temporal context to the model without needing recurrent layers. Table 5 shows the detection metrics for window sizes of 1, 6, and 12. The metrics improve as the window size increases, and hence a window of 12 timestamps is chosen (6 hours). However, with further increases in size, the metric gains diminish, and a longer window may not provide improvements that justify an exponential increase in model parameters.

Table 5: Ablation: window size.

Window	Acc.	Prec.	Rec.	F1
$W = 1$	0.8014	0.9195	0.4640	0.6168
$W = 6$	0.8271	0.9362	0.5344	0.6804
$W = 12$	0.8323	0.9009	0.5767	0.7033

Temporal columns

Table 6 shows that the *sin_hour* and *cos_hour* columns provide an improved model with better recall, F1, and accuracy for a minor trade-off in precision.

Table 6: Ablation: temporal columns.

Setting	Acc.	Prec.	Rec.	F1
Without temporal	0.8258	0.9057	0.5518	0.6858
With temporal	0.8323	0.9009	0.5767	0.7033

Physics-Aware mechanism

How physics is integrated into the model can be categorized into two types:

Loss-Only: The Physics Loss is added to the classification loss, but the leak head receives no features from the predicted flow and pressure.

Coupling: Along with the Physics Loss, the leak head receives 5 additional numerical features (mean, std, range) derived from the flow and pressure prediction head.

Table 7 shows that the model with coupling provides an increase of 0.02 in F1 over the loss-only model.

Table 7: Ablation: physics-aware mechanism (100 scenarios).

Variant	Description	F1
Loss only	Physics loss only	0.70
Coupling	pressure/flow features in leak head	0.72

Training Parameters

Table 8: Training hyperparameters for ANN and Physics-Aware ANN models.

Parameter	ANN	Physics-Aware ANN
Epochs	30	30
Batch Size	256	256
Learning Rate	1×10^{-3}	1×10^{-3}
Weight Decay	1×10^{-3}	1×10^{-3}
Optimizer	Adam	Adam
LR Scheduler	ReduceLROnPlateau	ReduceLROnPlateau
Random Seed	42	42
Dropout	0.5	0.5

Table 8 summarizes the training hyperparameters used for both the ANN and Physics-Aware ANN models.